

Lab Task-02

1. Write a script that asks the user for their name, program, batch, and course title, then saves this information into a text file `student_info.txt`.
2. Write a script that checks the current logged-in user and prints a personalized greeting with the current date and time.
3. Write a script that takes a directory name as input and creates a compressed backup (`tar.gz`) file with today's date in the filename.
4. Write a script that takes two numbers as input and performs addition, subtraction, multiplication, and division based on user choice. At the end, save the result into a file (e.g., `result.txt`) and display it using `cat`.
5. Write a script that generates numbers from 1 to 50, then saves even numbers into `even.txt` and odd numbers into `odd.txt`.
6. Write a script that asks the user for a project name and automatically creates folders: `src`, `bin`, and `docs` inside it.
7. Write a script that asks the user for a time (in minutes) and schedules a shutdown, printing a reminder message before execution.
8. Write a script with a menu using case that gives the user 4 options:
 - a) Show current date & time
 - b) List all files in current directory
 - c) Show current logged-in users
 - d) Exit
9. Write a script that goes through all `.txt` files in the current directory and renames them by adding today's date at the end (e.g., `report.txt` → `report_2025-08-29.txt`).
10. Write scripts /commands / syntax for moving files into three subdirectories directories:
 - a. `shelldir`, `cdir`, `jpgdir` according to their extensions.

Bonus Tasks Optional

Note: Questions Below are Of Weightage 05 In Karachi Campus OS Lab Mid Exam so it's for practice and completely optional

Task-1

Imagine you are tasked with monitoring a directory for new files and automatically processing them. You want to create a shell script that will do the following:

1. Monitor the directory for new files (Use a test directory or create one).
2. When a new file is detected, move it to a processing directory (Create one before running the line).

- 3.Process the file (in this case, let's say the processing involves converting a .C file to an output file).
- 4.Move the processed file to an output directory.
- 5.Delete the original file from the processing directory.

Write a shell script that accomplishes these tasks and can be run as a background process. Your script should also handle errors gracefully and log any issues to a file.

Task-02

Imagine you are tasked with automating the testing of a set of C and C++ programs on a Linux server. You want to create a shell script that will do the following:

- 1 .Find all C and C++ source code files in a specified directory and its subdirectories.
2. Compile each source code file into an executable file with a corresponding name.
- 3 .For each compiled executable file, run a set of predefined tests and output the results to a log file.
4. If any tests fail, print an error message to the console and continue to the next program.
5. If all tests pass, move the compiled executable file to a specified directory.

Write a shell script that accomplishes these tasks and can be run on the Linux server. Your script should also handle errors gracefully and log any issues to a file.